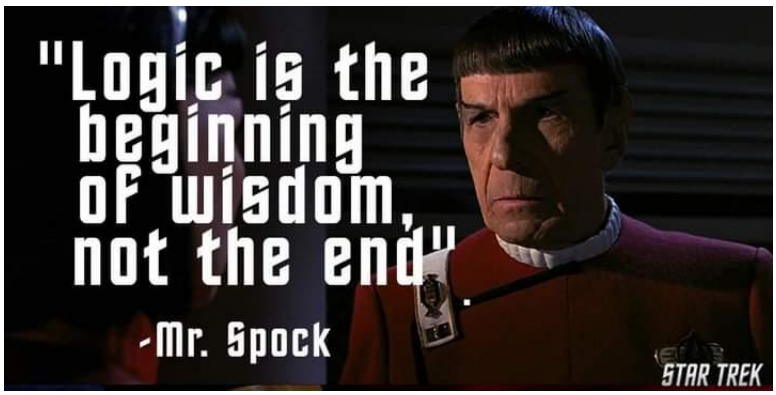
Chapter 1 - Logic and Proof

Note 1

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To understand mathematics and mathematical arguments, it is necessary to have a solid understanding of logic and the way in which known facts can be combined to prove new facts. Although many people consider themselves to be logical thinkers, the thought patterns developed in everyday living are only suggestive of and not totally adequate for the precision required in mathematics. In this chapter we take a careful look at the rules of logic and the way in which mathematical arguments are constructed. Note 1 presents the logical connectives that enable us to build compound statements from simpler ones. Note 2 discusses the role of quantifiers. Notes 3 and 4 analyze the structure of mathematical proofs and illustrate the various proof techniques by means of examples.



1.1 : Logical Connectives

The language of mathematics consists primarily of declarative sentences. If a sentence can be classified as true or false, it is called a **statement**. The truth or falsity of a statement is known as its truth value. For a sentence to be a statement, it is not necessary that we actually know whether it is true or false, but it must clearly be the case that it is one or the other.

Example 1 : Consider the following sentences.

(a). Two plus two equals four.

(b). Every continuous function is differentiable.

(c). $x^2 - 5x + 6 = 0$.

(d). A circle is the only convex set in the plane that has the same width in each direction.

(e). Every even number greater than 2 is the sum of two primes.

Solution :

Sentences (a) and (b) are statements since (a) is true and (b) is false.

Sentence (c), on the other hand, is true for some x and false for others. If we have a particular context in mind, then (c) will be a statement. In Note 2 we shall see how to remove this ambiguity.

Sentences (d) and (e) are more difficult. You may or may not know whether they are true or false, but it is certain that each sentence must be one or the other. Thus, (d) and (e) are both statements. [It turns out that (d) can be shown to be false, and the truth value of (e) has not yet been established.]

Example 2 : Which of the sentences are statements?

(a). If x is a real number, then $x^2 \geq 0$.

(b). Seven is a prime number.

(c). Seven is an even number.

(d). This sentence is false.

Solution : (a), (b), and (c)

In studying mathematical logic we shall not be concerned with the truth value of any particular simple statement. To be a statement, it must be either true or false (and not both), but it is immaterial which condition applies. What will be important is how the truth value of a compound statement is determined by the truth values of its simpler parts.

In everyday English conversation we have a variety of ways to change or combine statements. A simple statement like *It is windy* can be negated to form the statement *It is* ***not*** *windy*.

The compound statement It is windy *and* the waves are high is made up of two parts: "It is windy" and "The waves are high." These two parts can also be combined in other ways.
For example,

It is windy *or* the waves are high.

If it is windy, *then* the waves are high.

It is windy *if and only if* the waves are high.

The italicized words above (*not, and, or, if . . . then, if and only if*) are called **sentential connectives**. Their use in mathematical writing is similar to (but not identical with) their everyday usage. To remove any possible ambiguity, we shall look carefully at each and specify its precise mathematical meaning.

Let p stand for a given statement. Then $\sim p$ (read *not* p) represents the logical opposite (**negation**) of p. When p is true, $\sim p$ is false; when p is false, $\sim p$ is true. This can be summarized in a truth table:

p	$\sim p$
T	F
F	T

where T stands for true and F stands for false.

Example 3 : Let p, q, and r be given as follows:

p : Today is Monday.

q : Five is an even number.

r : The set of integers is countable.

Then their negations can be written as:

$\sim p$: Today is not Monday.

$\sim q$: Five is not an even number.

or

Five is an odd number.

$\sim r$: The set of integers is not countable.

or

The set of integers is uncountable.

The connective *and* is used in logic in the same way as it is in ordinary language. If p *and* q are statements, then the statement p and q (called the **conjunction** of p and q and denoted by $p \wedge q$) is true only when both p and q are true, and it is false otherwise.

Example 4 : Complete the truth table for $p \wedge q$. Note that we have to use four lines in this table to include all possible combinations of truth values of p and q.

Solution :

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

The connective or is used to form a compound statement known as a **disjunction**. In common English the word *or* can have two meanings. In the sentence;

We are going to paint our house yellow or green.

the intended meaning is *yellow or green, but not both.* This is known as the exclusive meaning of *or*. On the other hand, in the sentence

Do you want cake or ice cream for dessert?

the intended meaning may include the possibility of having both. This inclusive meaning is the *only* way the word *or* is used in logic. Thus, if we denote the disjunction p or q by $p \vee q$, we have the following truth table:

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

A statement of the form "If p, then q" is called an **implication** or a *conditional* statement. The if-statement p in the implication is called the *antecedent* and the then-statement q is called the *consequent*. To decide on an appropriate truth table for implication, let us consider the following sentence:

If it stops raining by Saturday, then I will go to the football game.

If a friend made a statement like this, under what circumstances could you call him a liar? Certainly, if the rain stops and he doesn't go, then he did not tell the truth. But what if the rain doesn't stop? He hasn't said what he will do then, so whether he goes or not, either is all right.

Although it might be argued that other interpretations make equally good sense, mathematicians have agreed that an implication will be called false only when the antecedent is true and the consequent is false. If we denote the implication "if p, then q" by $p \Rightarrow q$, we obtain the following table:

p	q	$p \Longrightarrow q$
T	T	T
T	F	F
F	T	T
F	T	T

It is important to recognize that in mathematical writing the conditional statement can be disguised in several equivalent forms. Thus the following expressions all mean exactly the same thing:

if p, then q	q provided that p
p implies q	q whenever p
p only if q	p is a sufficient condition for q
q if p	q is a necessary condition for p

Example 5 : Identify the antecedent and the consequent in each of the following statements.

(a). If n is an integer, then $2n$ is an even number.

(b). You can work here only if you have a college degree.

(c). The car will not run whenever you are out of gas.

(d). Continuity is a necessary condition for differentiability.

Solution :

(a). Antecedent: n is an integer
Consequent: $2n$ is an even number

(b). Antecedent: you can work here
Consequent: you have a college degree

(c). Antecedent: you are out of gas
Consequent: the car will not run

(d). Antecedent: differentiability
Consequent: continuity

One way to visualize an implication $R \Rightarrow S$ is to picture two sets R and S, with R inside S. Figure 1 shows several objects of different shapes. Some are round and some are not round. Some are solid and some are not solid. Objects that are round are in set R and objects that are solid are in set S. We see that the relationship between R and S in Figure 1 can be stated in several equivalent ways:

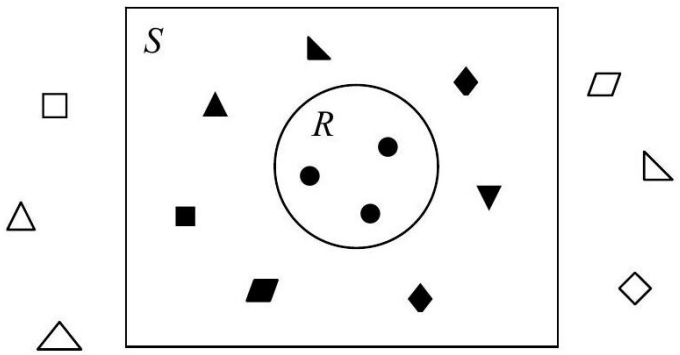


Figure 1: $R \Longrightarrow S$

- If an object is round (R), then it is solid (S).
- An object is solid (S) whenever it is round (R).
- An object is solid (S) provided that it is round (R).
- Being round (R) is a sufficient condition for an object to be solid (S). (It is sufficient to know that an object is round to conclude that it is solid.)
- Being solid (S) is a necessary condition for an object to be round (R). (It is necessary for an item to be solid in order for it to be round.)

Example 6 : In Figure 1, which of the following is correct?

(a). An object is solid (S) only if it is round (R).

(b). An object is round (R) only if it is solid (S).

Solution : Statement (b) is correct. If one of the objects is not solid, then it cannot possibly be round.

The statement "p if and only if q" (p iff q) is the conjunction of the two conditional statements $p \Longrightarrow q$ and $q \Longrightarrow p$. A statement in this form is called a **biconditional** and is denoted by $p \Longleftrightarrow q$.

p	q	$p \Longrightarrow q$	$q \Longrightarrow p$	$(p \Longrightarrow q) \wedge (q \Longrightarrow p)$
T	T	T	T	T
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T

Thus we see that $p \Longleftrightarrow q$ is true precisely when p and q have the same truth values.

Example 7 : Construct a truth table for each of the following compound statements.

$$\text{(a).}\ \sim (p \wedge q) \iff [(\sim p) \vee (\sim q)]$$

$$\text{(b).}\ \sim (p \vee q) \iff [(\sim p) \wedge (\sim q)]$$

$$\text{(c).}\ \sim (p \Rightarrow q) \iff [p \wedge (\sim q)]$$

Solution : Sometimes we condense a truth table by writing the truth values under the part of a compound expression to which they apply.

(a).

p	q	$\sim (p \vee q)$	$\iff$	$[(\sim p) \wedge (\sim q)]$
T	T	F T	T	F F F
T	F	T F	T	F T T
F	T	T F	T	T T F
F	F	T F	T	T T T

(b).

p	q	$\sim (p \vee q)$	$\iff$	$[(\sim p) \wedge (\sim q)]$
T	T	F T	T	F F F
T	F	F T	T	F F T
F	T	F T	T	T F F
F	F	T F	T	T T T

(c).

p	q	$\sim (\ p \vee q\ )$	$\iff$	$[\ \ p \wedge (\sim q)\ \ ]$
T	T	F T	T	T F F
T	F	T F	T	T T T
F	T	F T	T	F F F
F	F	F T	T	F F T

In Example 7 we find that each of the compound statements is true in all cases. Such a statement is called a **tautology**. When a biconditional statement is a tautology, it shows that the two parts of the biconditional are **logically equivalent**. That is, the two component statements have the same truth tables.

We shall encounter many more tautologies in the next few sections. They are very useful in changing a statement from one form into an equivalent statement in a different (one hopes simpler) form. In Example 7 (a) we see that the negation of a conjunction is logically equivalent to the disjunction of the negations. Similarly, in Example (b) we learn that the negation of a disjunction

is the conjunction of the negations. In Example (c) we find that the negation of an implication is not another implication, but rather it is the conjunction of the antecedent and the negation of the consequent.

Example 8 : Using Example 7 (a), we see that the negation of

The set S is compact and convex.

can be written as

The set S is not compact or it is not convex.

This example also illustrates that using equivalent forms in logic does not depend on knowing the meaning of the terms involved. It is the form of the statement that is important. Whether or not we happen to know the definition of "compact" and "convex" is of no consequence in forming the negation above.

Example 9 : Use the tautologies in Example 7 to write out a negation of each of the following statements.

(a). Seven is prime or $2 + 2 = 4$.

(b). If M is bounded, then M is compact.

Solution :

(a). Seven is not prime and $2 + 2 \neq 4$.

(b). M is bounded and M is not compact.

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